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EDUCATION

- National Yang Ming Chiao Tung University** Hsinchu, Taiwan
• *B.S. in Computer Science; GPA: 4.23/4.3 (Overall), 4.26/4.3 (Major)* Jul. 2021 - Jun. 2025
Ranking: 10/184 (4.42%)
- ETH Zurich** Zurich, Switzerland
• *Exchange Student in the Department of Computer Science* Sep. 2024 - Feb. 2025
Courses: Computer Graphics (6.0/6.0), Visual Computing (6.0/6.0), Computer Vision (5.5/6.0)

EXPERIENCES

- Google** Pixel Camera, On-device Perception, Computational Photography
• *Software Engineer* Sep. 2025 - Present
Software Engineer on the Pixel Camera Team, working on on-device perception and computational photography.
- CompPhoto Lab, NYCU** Python, Research, NeRF, 3DGS, Diffusion Models, Embodied AI
• *Research Assistant and System Administrator* Aug. 2023 - Present
Advised by Prof. Yu-Lun Liu. Working on 3D scene synthesis, generative models for vision, and embodied AI, with a focus on NeRF, 3D Gaussian Splatting, and vision-language navigation. Led Skyfall-GS (ECCV 2026), which synthesizes immersive 3D urban scenes from satellite imagery. Continued as a research assistant after receiving my B.S. (Sep. 2025).
- Google** SAM, Research, Knowledge Distillation
• *Software Engineering Intern* Jun. 2024 - Sep. 2024
During my internship with the Pixel Camera Team, I spearheaded the integration of the Segment Anything Model (SAM) into mobile devices. My work involved conducting a comprehensive review of cutting-edge research papers to identify optimal strategies for deploying complex models on mobile platforms. Leveraging knowledge distillation techniques, I successfully transferred SAM's capabilities to a lightweight MobileNetV4 architecture. This resulted in remarkable outcomes: achieving only a 4% performance drop compared to the original model, while significantly reducing latency to 20 ms (8x faster) on a mobile TPU.
- IT Center of CS Department, NYCU** Linux, Kubernetes, PHP, Laravel, Vue, CI/CD, GitLab flow
• *Teaching Assistant of WWW and System Group* Jul. 2022 - Jun. 2025
Developed the backend of teaching assistant contact information system using PHP and Laravel. And learned how to design the database schema and write RESTful API, also used OpenAPI to write API document. Also be familiar with the process of software development, including git version control, merge requests and CI/CD.
- Microsoft** AzureML, NLP, Address Parsing, C#, Aether, jQuery
• *Research & Development Intern at Bing Geocoding* Sep. 2023 - Jun. 2024
Integrated code in backend component, made it able to populate new data format and visualized them in dashboard. Validated the DeepCAL ML model and improved the training pipeline.
- Appier** MongoDB, Trino, Argo Workflow, Kubernetes, GCP, BigQuery, Looker Studio, Agile, System Design
• *Backend Engineer Summer Intern* Jul. 2023 - Sep. 2023
Developed a novel system to calculate and visualize the cloud service costs for each customer per feature, which replaced the traditional manual estimation. By automatically collecting and calculating cost, it achieving 7200x speedup. Utilized Google Cloud Storage and Looker Studio for visualization. Developed in collaboration with the team's Scrum process. Also engaged with PMs and sales teams to ensure it meets their expectations.

PUBLICATIONS

- Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery:**
Jie-Ying Lee, Yi-Ruei Liu, Shr-Ruei Tsai, Wei-Cheng Chang, Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, Yu-Lun Liu
European Conference on Computer Vision (ECCV), 2026.
Skyfall-GS synthesizes photorealistic, city-block scale 3D urban scenes from satellite imagery using diffusion models, eliminating the need for expensive 3D scanning and manual annotation while enabling real-time exploration.

- **Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion:**

Ting-Hsuan Chen*, Ying-Huan Chen*, Tao Tu, **Jie-Ying Lee**, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo†, Yu-Lun Liu†, Yue Wang†, Liu Ren (* denotes equal contribution, † denotes corresponding author)

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

Pantheon360 is a controllable 360° video diffusion framework for digital twin generation that synthesizes high-fidelity videos from sparse 360° inputs, letting the diffusion model focus on photorealistic texture refinement while a 3D Cache enforces global geometric consistency.

- **BRDFusion: Physics Meets Generation for Urban Scene Inverse Rendering:**

Yi-Ruei Liu, **Jie-Ying Lee**, Zheng-Hui Huang, Yu-Lun Liu†, Chih-Hao Lin† († denotes corresponding author)

arXiv preprint, 2026.

BRDFusion is an inverse rendering framework for urban scenes that unifies physics-based rendering with generative diffusion priors to recover explicit material and lighting properties from video, enabling high-quality relighting and simulation.

- **LightsOut: Diffusion-based Outpainting for Enhanced Lens Flare Removal:**

Shr-Ruei Tsai, Wei-Cheng Chang, **Jie-Ying Lee**, Chih-Hai Su, Yu-Lun Liu

IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

LightsOut is a diffusion-based outpainting framework that enhances lens flare removal by reconstructing off-frame light sources, combining a multitask regression module with LoRA fine-tuned diffusion models to produce realistic and physically consistent results.

- **See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation:**

Chih-Yao Hu*, Yang-Sen Lin*, Yuna Lee, Chih-Hai Su, **Jie-Ying Lee**, Shr-Ruei Tsai, Chin-Yang Lin, Kuan-Wen Chen, Tsung-Wei Ke, Yu-Lun Liu (* denotes equal contribution)

Conference on Robot Learning (CoRL), 2025.

See, Point, Fly (SPF) is a training-free framework for aerial vision-and-language navigation. By leveraging vision-language models and reformulating navigation as a 2D spatial grounding task, SPF enables universal unmanned aerial navigation without task-specific training.

- **AuraFusion360: Augmented Unseen Region Alignment for Reference-based 360° Unbounded Scene Inpainting:**

Chung-Ho Wu*, Yang-Jung Chen*, Ying-Huan Chen, **Jie-Ying Lee**, Bo-Hsu Ke, Chun-Wei Tuan Mu, Yi-Chuan Huang, Chin-Yang Lin, Min-Hung Chen, Yen-Yu Lin, Yu-Lun Liu (* denotes equal contribution)

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

The approach introduces (1) depth-aware unseen mask generation for accurate occlusion identification, (2) Adaptive Guided Depth Diffusion, a zero-shot method for accurate initial point placement without requiring additional training, and (3) SDEdit-based detail enhancement for multi-view coherence.

- **SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes:**

Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu, **Jie-Ying Lee**, Jiun-Long Huang, Yu-Chee Tseng, Yu-Lun Liu

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

SpectroMotion is a novel approach that combines 3D Gaussian Splatting with physically-based rendering (PBR) and deformation fields to reconstruct dynamic specular scenes and is the only existing 3DGS method capable of synthesizing photorealistic real-world dynamic specular scenes.

- **BoostMVSNeRFs: Boosting MVS-based NeRFs to Generalizable View Synthesis in Large-scale Scenes:**

Chih-Hai Su*, Chih-Yao Hu*, Shr-Ruei Tsai*, **Jie-Ying Lee***, Chin-Yang Lin, Yu-Lun Liu (* denotes equal contribution)

ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH), 2024.

BoostMVSNeRFs aims to enhance the rendering quality of MVS-based NeRFs in large-scale scenes. Our method significantly improves the rendering quality in large-scale scenes and unbounded outdoor scenes.

ACADEMIC SERVICE

- **Reviewer:** ECCV, IJCV

- **Teaching Assistant:** Image and Video Generation, NYCU (2025 Fall)

PROJECTS

- **Physically-based Renderer (ETH Zurich Computer Graphics):** Extended the Nori educational ray tracer in C++ through four programming assignments and a final project. Assignments covered normal and average visibility integrators, sampling and warping with chi-square validation, Monte Carlo direct illumination with emitter sampling, and global illumination with path tracing (BSDF sampling and multiple importance sampling) and a dielectric BSDF. In the two-person final project (with Wei Wang), independently designed and implemented heterogeneous participating media with single- and multiple-scattering volumetric path tracers, equiangular sampling of single scattering, the Henyey-Greenstein anisotropic phase function, directional light and environment map emitters, a windowed-sinc reconstruction filter, and Blender-to-Nori scene modeling, rendering a photorealistic waterfall cave scene. Selected for the final round of the course rendering competition with an in-person presentation. (Dec. 2024)
- **Operating System Capstone: Bare-metal Kernel on Raspberry Pi:** Implemented a bare-metal OS kernel in C and assembly for Raspberry Pi 3B+ (ARMv8-A, AArch64) across eight labs: UART shell and mailbox interface, a self-hosted UART bootloader with device tree parsing, exception and interrupt handling, a buddy-system memory allocator, threads and user processes with system calls and signals, virtual memory, and a virtual file system with tmpfs, initramfs, and device file systems. [Link](#) (Jun. 2024)

- **P Language Compiler (NYCU Compiler Design):** Built a complete compiler in C++ for P, a Pascal-like language, using flex and bison: scanner, parser, AST construction, semantic analysis with symbol tables, and RISC-V code generation executed on the Spike simulator. [Link](#) (Jan. 2024)
- **VAE-pix2pix-terrain-generator:** Used NASA's SRTM 1 Arc-Second dataset to collect altitude maps and MapTiler to collect corresponding satellite images from around the world. The collected images are used to train a VAE-pix2pix model, which is an VAE combined with pix2pix. The model adds realistic details to the heightmap and generates corresponding satellite images. [Link](#) (Jul. 2020)
- **C++ Implementation of Root-parallelization MCTS:** Project for theory of computer games. Implemented a root-parallelization multi-thread MCTS algorithm for NoGo game. It can be also applied in various of games, like Go, Hex, etc. [Link](#) (Jan. 2023)

AWARDS

- **Shun-I Chu Zyxel Scholarship:** Awarded to one of only two computer science students for exceptional academic performance.
- **NSTC Research Grant for University Students:** Awarded the undergraduate research grant by the National Science and Technology Council, Taiwan (2024).
- **Third Place of the NYCU CS Undergraduate Research Competition:** Awarded for the project "BEVGaussian: Generate Scene-level 3D Gaussian from BEV image."
- **Bronze Award of The 2022 ICPC Asia Taoyuan Regional Programming Contest:** Thirty-second place out of 100 teams. Second place of the bronze award.
- **Academic Achievements Award in 2021 Fall Semester:** Top 5% in the class.
- **Fundamental Course Award of Discrete Mathematics, Digital Circuit Design, Data Structures & Object-oriented Programming and Introduction to Algorithms:** Top 5% of these courses.
- **Marconix Award of 19th Marconix Science Award:** Top award in Marconix Science Award. The only recipient within ten years.
- **Second Place in 60th National Primary and High School Science Fair:** Awarded the best work at the 60th national science fair (The first place is absent) out of 8 all participants who are the first place from cities in Taiwan.
- **Fourth Prize in 2021 Taiwan International Science Fair:** Be awarded the fourth prize out of 21 participants from world.